

PRODUCT OPTIMIZATION & REVENUE CONTRIBUTION ANALYSIS

Afficionado Coffee Roasters

Internship Research Report

Unified Mentor Program

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Abstract

This research paper presents a comprehensive product-level performance analysis for Afficionado Coffee Roasters, a specialty coffee retailer operating across multiple store locations. The study analyzes 149,116 transaction records to identify revenue drivers, product popularity patterns, and menu optimization opportunities. Using Pareto analysis, quadrant classification, and category-level revenue decomposition, this report uncovers that 42 out of 80 products drive 80% of total revenue, Coffee and Tea together account for 66.7% of revenue, and Barista Espresso is the single highest-revenue product type. Actionable recommendations are provided for menu simplification, hero product protection, and pricing strategy improvements to maximize profitability.

1. Introduction

1.1 Background

In the competitive specialty coffee retail industry, understanding product-level performance is critical for sustainable profitability. Many coffee shops maintain extensive menus under the assumption that more variety attracts more customers. However, this approach often leads to operational inefficiencies, slower service, and diluted focus on truly profitable products.

1.2 Problem Statement

Afficionado Coffee Roasters collects transaction-level data across all its store locations but lacks structured analytics to answer critical business questions:

- Which products are truly driving revenue vs. merely selling in volume?
- Which product categories represent the highest business dependency?
- Which menu items are underperforming and candidates for removal?
- How concentrated is revenue across the menu portfolio?

1.3 Objectives

Primary objectives of this analysis:

- Identify top-selling and least-selling products by volume and revenue
- Quantify revenue contribution by product and category
- Measure revenue concentration across the menu using Pareto analysis
- Classify products into performance quadrants for strategic decision-making

2. Dataset Description

The dataset used in this analysis consists of transaction records from Maven Roasters, a fictitious coffee shop operating across three New York City locations. The dataset contains 149,116 rows and 11 columns covering the period from January to June 2023.

Column	Description	Data Type
transaction_id	Unique identifier per transaction	Integer
transaction_date	Date of transaction (MM/DD/YY)	Date
transaction_time	Time of transaction (HH:MM:SS)	Time
transaction_qty	Quantity purchased per transaction	Integer
unit_price	Price per unit (\$)	Float
store_id	Store identifier	Integer
store_location	Physical store location	String
product_id	Unique product identifier	Integer
product_category	Broad product group	String
product_type	Product variant within category	String
product_detail	Detailed product attributes	String

2.1 Data Quality Assessment

A thorough data validation was performed prior to analysis. The dataset demonstrated excellent quality with no issues requiring remediation:

- Zero null or missing values across all 11 columns
- Zero negative or zero values in transaction_qty and unit_price
- All product identifiers and categories were consistent and valid
- Revenue computation was straightforward: Revenue = transaction_qty x unit_price

3. Analytical Methodology

3.1 Feature Engineering

A new revenue column was engineered at the transaction level as the product of transaction quantity and unit price. This derived metric serves as the foundation for all subsequent analyses, enabling comparison between product popularity (volume-based) and product profitability (revenue-based).

3.2 Product Popularity Analysis

Products were grouped by product_detail and aggregated by total units sold (transaction_qty). Rankings were generated from highest to lowest to identify the top and bottom performing products by sales volume. This analysis answers the question: What do customers buy most?

3.3 Revenue Contribution Analysis

Products were grouped by product_detail and aggregated by total revenue. A revenue share percentage was computed for each product as its proportion of total portfolio revenue. This analysis answers: What products generate the most money?

3.4 Category & Product-Type Analysis

Revenue was aggregated at two levels: broad product category (Coffee, Tea, Bakery, etc.) and product type within each category (Barista Espresso, Brewed Chai Tea, etc.). Revenue share percentages were computed at each level to identify category dependency and within-category performance leaders.

3.5 Pareto (80/20) Analysis

A cumulative revenue share percentage was computed across all products sorted by descending revenue. The Pareto threshold was set at 80%, identifying the minimum number of products required to account for 80% of total portfolio revenue. This analysis quantifies menu concentration risk and identifies revenue anchors.

3.6 Popularity vs Revenue Quadrant Analysis

Products were mapped onto a two-dimensional matrix using units sold (x-axis) and revenue (y-axis). Median values of both dimensions were used as classification thresholds. Each product was assigned to one of four strategic quadrants:

- Hero Products: Above median in both units sold and revenue
- Hidden Gems: Below median in units sold but above median in revenue
- High Volume Low Revenue: Above median in units sold but below median in revenue
- Dead Weight: Below median in both units sold and revenue

4. Analysis & Findings

4.1 Product Popularity Analysis

The top 10 products by units sold are dominated by Tea and Coffee variants, with all top performers registering between 4,000 and 4,700 units sold. The narrow range across the top 10 indicates relatively uniform popularity among the best-selling products.

Key finding: Earl Grey Rg leads all products in units sold with approximately 4,708 units, followed by Dark Chocolate Lg and Morning Sunrise Chai Rg. Notably, the top-selling product by volume does not appear in the top 10 by revenue — indicating a significant popularity-profitability gap.

Rank	Product	Units Sold
1	Earl Grey Rg	4,708
2	Dark Chocolate Lg	4,668
3	Morning Sunrise Chai Rg	4,643
4	Latte	4,602
5	Peppermint Rg	4,564

4.2 Revenue Contribution Analysis

Revenue rankings reveal a different picture from popularity rankings. Sustainably Grown Organic Lg leads all products with \$21,151 in total revenue, followed by Dark Chocolate Lg at \$21,006. The top revenue-generating products tend to be large-sized premium items, reflecting the impact of unit price on total revenue.

Rank	Product	Total Revenue	Revenue Share %
1	Sustainably Grown Organic Lg	\$21,152	3.03%
2	Dark Chocolate Lg	\$21,006	3.01%
3	Latte Rg	\$19,112	2.73%
4	Cappuccino Lg	\$17,642	2.52%
5	Morning Sunrise Chai Lg	\$17,384	2.49%

4.3 Category Revenue Analysis

Revenue distribution across product categories reveals a strong concentration in Coffee and Tea:

Category	Revenue Share	Strategic Importance
Coffee	38.6%	Primary revenue driver
Tea	28.1%	Strong secondary driver
Bakery	11.8%	Solid complementary category
Drinking Chocolate	10.4%	Meaningful contributor
Coffee Beans	5.7%	Low but targeted audience
Other Categories	5.4%	Minimal combined impact

Coffee and Tea combined account for 66.7% of total revenue, representing a significant category concentration. While these are the core offerings of a coffee shop, the heavy dependency on two categories represents a business risk if either category faces disruption.

4.4 Product Type Analysis

At the product type level, Barista Espresso dominates all other types with approximately \$90,000 in revenue, nearly 25% more than the second-ranked type, Brewed Chai Tea at \$75,000. This makes Barista Espresso the single most critical product type in the entire portfolio.

4.5 Pareto Analysis

The Pareto analysis reveals that 42 out of 80 products (52.5%) are required to reach 80% of total revenue. While slightly above the classic 20/80 rule, this still represents significant revenue concentration:

- 42 products drive 80% of all revenue
- The remaining 38 products contribute only 20% of revenue
- The Pareto curve shows a steep initial rise followed by a long flat tail — confirming that a large portion of the menu has minimal financial impact

This finding has direct menu optimization implications: removing or redesigning the bottom 38 products could simplify operations with minimal revenue impact.

4.6 Popularity vs Revenue Quadrant Analysis

The scatter plot matrix reveals the strategic positioning of all 80 products across four performance quadrants:

Quadrant	Characteristics	Strategic Action
Hero Products	High units sold + High revenue	Protect, promote, never remove
Hidden Gems	Low units sold + High revenue	Increase visibility and marketing
High Vol Low Revenue	High units sold + Low revenue	Review pricing strategy
Dead Weight	Low units sold + Low revenue	Consider removal from menu

The majority of products fall into either the Hero or Dead Weight quadrants, with very few Hidden Gems identified. This suggests that pricing strategy for underexposed high-value products could be improved to surface more revenue from lower-volume items.

5. Recommendations

5.1 Protect Hero Products

Hero products represent the core of Afficionado's revenue engine. These products should never be removed from the menu, should always be well-stocked, and should receive prominent placement on the menu board. Staff training should prioritize knowledge of these items to maximize upselling opportunities.

5.2 Promote Hidden Gems

Hidden Gem products generate strong revenue despite low sales volume. These products are likely underexposed or unknown to customers. Recommendations include featuring them as daily specials, training baristas to suggest them proactively, and highlighting them on digital menu boards.

5.3 Review Dead Weight Products

The 38 products outside the top 42 revenue contributors should be reviewed for potential removal or redesign. Removing low-performing items will simplify the menu, reduce inventory costs, speed up service, and allow staff to focus on high-value products. A phased removal approach is recommended to monitor customer response.

5.4 Address Category Concentration Risk

The 66.7% revenue dependency on Coffee and Tea represents a concentration risk. Afficionado should develop strategies to grow the Bakery and Drinking Chocolate categories, which show healthy contribution levels. Seasonal promotions, bundle deals, and cross-selling with coffee products can help diversify revenue streams.

5.5 Repricing Strategy for High Volume Low Revenue Products

Products with high sales volume but low revenue are leaving money on the table. A modest price increase of 5-10% on these items could significantly improve their revenue contribution while retaining most customers due to their established popularity.

6. Conclusion

This analysis successfully delivers product-level intelligence for Afficionado Coffee Roasters by shifting focus from transactional volume to revenue contribution. The key findings are clear: 42 products drive 80% of revenue, Coffee and Tea dominate category performance, Barista Espresso is the single highest-value product type, and a significant portion of the menu contributes minimally to total revenue.

The Pareto analysis and quadrant classification provide a structured framework for menu optimization decisions. By acting on the recommendations in this report, Afficionado Coffee Roasters can simplify its menu, protect its revenue anchors, surface hidden value in underperforming gems, and reduce business risk through category diversification.

This analysis was conducted using Python (pandas, plotly) for data processing and visualization, and an interactive Streamlit dashboard was developed to enable ongoing self-service analytics for business stakeholders.

7. References & Tools

- Dataset: Coffee Shop Sales Dataset
- Python 3.x — Data processing and analysis
- Pandas — Data manipulation and aggregation
- Plotly Express — Interactive data visualization
- Streamlit — Interactive dashboard development
- Jupyter Notebook — Exploratory data analysis environment